

Exp. No. : 2

Exp. Name: Noise Figure and SNR Degradation.

Aim: To study and analyze the effect of noise figure on the degradation of Signal-to-Noise Ratio (SNR) in a communication system using MATLAB.

Introduction:

In communication and electronic systems, maintaining signal quality is essential for reliable performance. One of the key parameters used to measure signal quality is the Signal-to-Noise Ratio (SNR), which compares the strength of a desired signal to the level of background noise.

However, real systems such as amplifiers and receivers introduce additional noise, leading to SNR degradation. This degradation is quantified using a parameter called the Noise Figure (NF), which indicates how much noise a system adds relative to an ideal system.

This experiment uses MATLAB to analyze and visualize how different noise figure values affect the output SNR of a system.

$$SNR_{out} = SNR_{in} - NF$$

The results are plotted to show how increasing noise figure leads to greater degradation in signal quality. An ideal case ($NF = 0$ dB) is also included for comparison

Procedure:

Step 1: Initialize the Environment

- Clear all variables, close figures, and clear the command window.

Step 2: Define Parameters

- Set a range of input SNR values from (0 dB - 30 dB).
- Define multiple noise figure values (3 dB, 6 dB, 10 dB, 15 dB).

Step 3: Create a Plot

- Open a new figure window.
- Enable multiple plots on the same graph using hold on.

Step 4: Compute Output SNR

- For each noise figure value:
 - Subtract NF from the input SNR to get output SNR.
 - Plot the resulting curve.

Step 5: Plot Ideal Case

- Plot a reference line where output SNR equals input SNR
(NF = 0 dB).

Step 6: Format the Graph

- Add grid, labels, title, and legend.

Discussion Questions:

1. Why does increasing NF lead to greater SNR degradation?
2. Why are all plotted curves parallel to the ideal line?
3. In real-world applications, why is it important to minimize noise figure?
4. How would the graph change if NF = 0 dB for all cases? Explain and draw the results.

MATLAB Code Used:

Experiment 5: Noise Figure and SNR Degradation

Analyze how noise figure affect system performance

```
clear all; close all; clc;

% Parameters
SNR_in_dB = 0:0.5:30;      % Input SNR in dB
NF_dB = [3, 6, 10, 15];   % Noise Figure values in dB

% Calculate output SNR for different noise figures
figure('Name', 'SNR Degradation due to Noise Figure');
hold on;
colors = {'b', 'r', 'g', 'm'};

for i = 1:length(NF_dB)
    SNR_out_dB = SNR_in_dB - NF_dB(i);
    plot(SNR_in_dB, SNR_out_dB, [colors{i}, '-'], 'Linewidth', 2);
end

% Plot ideal case (no degradation)
plot(SNR_in_dB, SNR_in_dB, 'k--', 'Linewidth', 1.5);

grid on;
xlabel('Input SNR (dB)'); ylabel('Output SNR (dB)');
title('SNR Degradation vs Noise Figure');
legend('NF = 3 dB', 'NF = 6 dB', 'NF = 10 dB', 'NF = 15 dB', 'Ideal (NF = 0 dB)');
```

Results:

